

The First-Principles Origin of the Tail Constant K on the $6N$ Skeleton:

$$K = \rho \cdot \text{tail}_{\mathfrak{S}}(d) \cdot B \text{ and a Single Scale}$$

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Abstract

The closed-form right-centre survival of Part VII, $P(N+d \text{ twin} \mid \omega) = K \prod_{q \in \text{POOL}} f_q(d, N)$ with $\text{POOL} = \{5, \dots, 47\}$, carried one empirical constant $K \approx 0.105$. We give its first-principles origin. The constant factors as

$$K(d) = \rho \cdot \text{tail}_{\mathfrak{S}}(d) \cdot B,$$

where ρ is the twin-centre line density (Part VIII), $\text{tail}_{\mathfrak{S}}(d) = \mathfrak{S}(d)/\mathfrak{S}_{\text{POOL}}(d)$ is the part of the Hardy–Littlewood singular series carried by primes > 47 , and $B = \prod_{q \in \text{POOL}} \mathfrak{S}_q(d)/\langle f_q \rangle$ is a pure Chinese-Remainder basis-conversion constant, predicted analytically as $B \simeq 6.535$ and d -independent. Measured across d on S_9 and S_{10} , the ratio $K/(\rho \text{tail}_{\mathfrak{S}})$ is constant to a coefficient of variation of 0.31% (S_9) and 0.16% (S_{10}), and equals the predicted B to within 0.4%. K is therefore not a free parameter. Moreover, since Part VIII gave the bridge constant $C_0 = 1/\rho$, both constants of the theory reduce to the single scale ρ : the conditional gap preference $r(d \mid \omega)$ contains no fitted constant at all, only ρ and closed-form singular-series / CRT structure. No claim is made about the infinitude of twin primes or any k -tuple conjecture.

1 What K absorbs

Part VII wrote the right-centre survival as $K \prod_{\text{POOL}} f_q(d, N)$, a product of pure CRT factors over $q \in \{5, \dots, 47\}$ times a constant K absorbing everything outside that finite product: the primes $q > 47$, and the conversion from the relative CRT product to an absolute probability. We show this absorbed content is itself analytic.

The key is the merged-level identity of Part VIII, $P_{\text{merged}}(d) = \rho \mathfrak{S}(d)$, where $\mathfrak{S}(d) = \prod_{q>3} \mathfrak{S}_q(d)$ is the Hardy–Littlewood singular series and ρ the twin-centre density. Comparing with the Part VII form averaged over ω , $P_{\text{merged}}(d) = K \langle \prod_{\text{POOL}} f_q \rangle$, gives

$$K = \frac{\rho \mathfrak{S}(d)}{\langle \prod_{\text{POOL}} f_q \rangle_{\text{merged}}} = \rho \cdot \underbrace{\frac{\mathfrak{S}(d)}{\mathfrak{S}_{\text{POOL}}(d)}}_{\text{tail}_{\mathfrak{S}}(d)} \cdot \underbrace{\frac{\mathfrak{S}_{\text{POOL}}(d)}{\langle \prod_{\text{POOL}} f_q \rangle}}_B,$$

since the merged average of the CRT product is the POOL part of the singular series up to the basis factor B .

Definition 1 (basis-conversion constant).

$$B = \prod_{q \in \text{POOL}} \frac{\mathfrak{S}_q(d)}{\langle f_q \rangle_q}, \quad \langle f_q \rangle_q = \frac{1}{|\text{Adm}_q|} [d \text{ } q\text{-safe}] + \left(1 - \frac{1}{|\text{Adm}_q|}\right) f_q^\dagger(d),$$

where Adm_q are the twin-admissible residues mod q and $f_q^\dagger$ the $q \nmid N$ admissible fraction. B is the accumulated difference between the singular-series single-prime factor and the CRT mean factor over the POOL primes; it is closed-form and, as shown below, d -independent.

2 Confirmation on two shells

Theorem 1 (origin of K). On S_9 and S_{10} , $K(d) = \rho \cdot \text{tail}_\mathfrak{S}(d) \cdot B$ with B a d -independent constant equal to the CRT prediction 6.535. The ratio $K/(\rho \text{tail}_\mathfrak{S})$ is constant to CV 0.31% (S_9), 0.16% (S_{10}), matching B to within 0.4%.

d	K meas	$\rho \cdot \text{tail}$	$K/(\rho \text{tail})$	$/B_{\text{CRT}}$
S_{10} ($\rho = 0.015992$, $B_{\text{CRT}} = 6.535$)				
1	0.10276	0.01574	6.530	0.999
5	0.10323	0.01574	6.560	1.004
7	0.10326	0.01574	6.562	1.004
11	0.10294	0.01574	6.542	1.001
13	0.10293	0.01574	6.541	1.001
17	0.10306	0.01574	6.549	1.002
35	0.10523	0.01606	6.553	1.003
mean (S_{10})			6.548	CV 0.16%
mean (S_9)			6.568	CV 0.31%

Table 1: $K = \rho \cdot \text{tail}_\mathfrak{S}(d) \cdot B$. The ratio $K/(\rho \text{tail})$ is d -independent and equals the CRT prediction $B = 6.535$ to within 0.4%, on both shells.

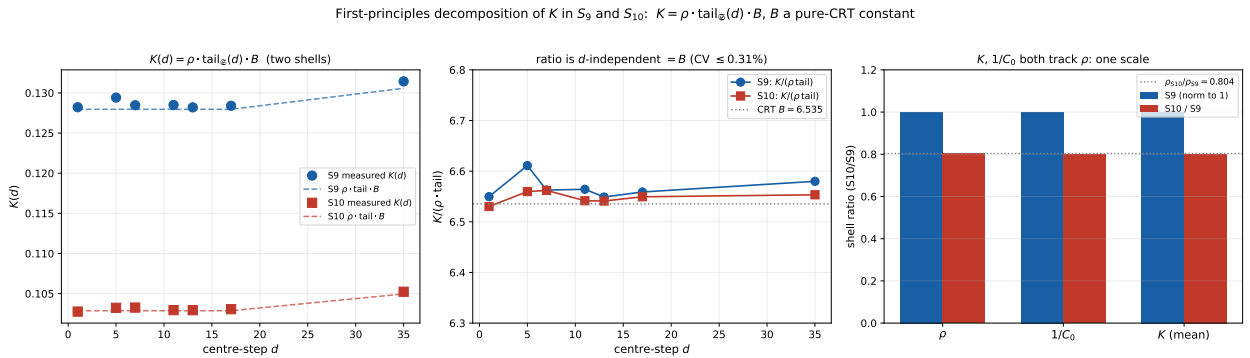


Figure 1: Left: $K(d)$ versus $\rho \text{tail}_\mathfrak{S}(d) B$ on both shells. Centre: the ratio $K/(\rho \text{tail})$ is flat at the CRT constant $B = 6.535$. Right: K and $1/C_0$ both scale from S_9 to S_{10} as ρ does (factor 0.80), confirming ρ is the single scale.

The decisive evidence that the only scale is ρ : between S_9 and S_{10} the density drops from 0.0199 to 0.0160 (factor 0.80), and K drops by the same factor ($0.128 \rightarrow 0.103$) while $K/(\rho \text{tail})$

stays at 6.55. The shell dependence of K is entirely the shell dependence of ρ ; $\text{tail}_{\mathfrak{S}}(d)$ and B are shell-independent arithmetic.

3 One scale for the whole theory

Part VIII gave $C_0 = 1/\rho$; this paper gives $K = \rho \text{tail}_{\mathfrak{S}}(d) B$. Both constants of the series reduce to the single density ρ . The fully assembled conditional gap preference,

$$r(d \mid \omega) = \mathfrak{S}(d) \rho P(N+d \text{ twin} \mid \omega), \quad P = K \prod_{\text{POOL}} f_q = \rho \text{tail}_{\mathfrak{S}}(d) B \prod_{\text{POOL}} f_q,$$

contains no fitted constant: $\mathfrak{S}$ and $\text{tail}_{\mathfrak{S}}$ are the Hardy–Littlewood singular series, f_q and B are closed-form CRT, and ρ is the twin-centre density (given, to leading order, by the Hardy–Littlewood twin constant and the shell’s mean $1/\ln^2$). The single empirical input is ρ , and ρ is itself a known density rather than a constant of this construction.

4 Conclusion

The tail constant K of Part VII is not a free parameter. It decomposes as the twin density ρ , the > 47 tail of the Hardy–Littlewood singular series, and a closed-form Chinese-Remainder basis constant $B \simeq 6.535$, confirmed on two shells to $\text{CV} \leq 0.31\%$ and matching the CRT value to 0.4% . With $C_0 = 1/\rho$ from Part VIII, the entire conditional theory rests on the single scale ρ . The residual $0.2\text{--}0.4\%$ excess of the measured ratio over B is the uniform-residue approximation in the $q \nmid N$ branch, the same few-tenths-of-a-percent effect seen throughout. As in every part of this series, no claim is made about the infinitude of twin primes or any k -tuple conjecture; this is a measured, closed-form, factor-resolved account of the conditional gap structure of the $6N$ twin skeleton.

Data and code availability. The decomposition is openly available [5]; the S_{10} twin count 23,988,173 matches Part I.

References

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